PROBLEM 26

# Heavy Particle on a Rotating Half-Circumference (Moving-Frame Method)

Original statement · Problems.pdf, p. 14

Rotating Wire + Inertia Forces

A heavy point particle $P$ of mass $m$ is constrained to move without friction along the half-circumference wire of radius $a$ shown in the figure. The wire containing the point particle rotates around the $OY$ axis with constant angular velocity $\boldsymbol{\omega} = \Omega\mathbf{j}$. Initially, the particle is located at $x = 0$ with velocity $v_0$ along the positive $x$ axis.

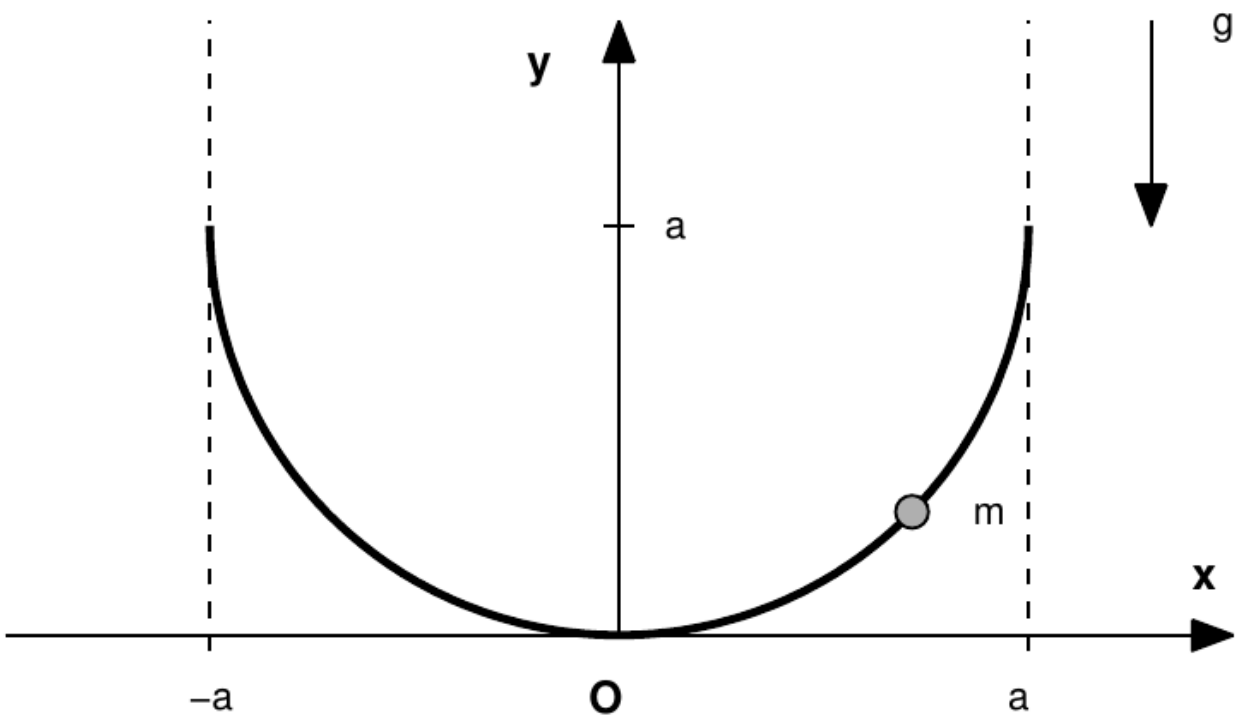



*Figure: Half-circumference wire of radius a with lowest point O, ends at x = ±a, rotating about the vertical axis OY; particle of mass m on the wire.*

**(a)** Determine the number of degrees of freedom of the problem.

**(b)** Define a convenient moving reference frame with origin in $O$, $OX_1Y_1Z_1$. Express the inertia force that acts on the particle in this moving reference frame.

**(c)** Express also the components of the weight of the particle with respect to $OX_1Y_1Z_1$.

**(d)** Determine if there are any equilibrium points.

**(e)** Reduce the problem to integrals (i.e. quadratures) and determine the value of the reaction force.

**(f)** Is it possible to use the energy equation? If $\Omega$ were a function of time, would it be possible to use the energy equation? Justify your answer.

**Worked Solution (4-Phase Methodology)**

PHASE 1

### Physical Statement, Hypotheses & Parameters

- **Physical system:** a heavy particle $P$ (mass $m$) slides without friction on a half-circumference wire of radius $a$. In the plane of the wire the lowest point is $O$, the centre is $C = (0, a)$ and the ends are $(\pm a, a)$ (figure). The wire turns about the vertical axis $OY$ with constant angular velocity $\boldsymbol{\omega} = \Omega\,\mathbf{j}$.
- **Hypotheses:** the frame $OXYZ$ with $OY$ vertical upwards is inertial; gravity is $-g\,\mathbf{j}$; the wire is smooth, so its reaction has no component along the wire; the rotation is imposed (whatever torque is needed is supplied). This problem was solved in Topic 3; here it is solved again with the relative-motion method of Topic 5 (moving frame, Notes Eq. 3.27 and inertia forces, Notes Eqs. 4.62–4.63).
- **Why a moving frame:** in a frame that turns with the wire the constraint is a fixed circle, so the motion of $P$ is a one-dimensional motion on a fixed curve. The cost is to add the centrifugal and Coriolis forces; there is no Euler force and no force from the acceleration of the origin because $\Omega$ is constant and $O$ is fixed.
- **Initial conditions ($t = 0$):** $P$ at $O$ ($x = 0$) with velocity $v_0$ along $+x$.

| Symbol | Meaning | SI unit |
|---|---|---|
| $m$ | mass of $P$ | kg |
| $a$ | radius of the wire | m |
| $\Omega$ | angular velocity of the wire (constant) | rad/s |
| $g$ | gravitational acceleration | m/s$^2$ |
| $v_0$ | initial speed at $O$ | m/s |
| $\theta$ | angle of $CP$ from the downward vertical | rad |

PHASE 2

### Reference Frames, Vector Bases & Rotation Matrices

- **Inertial frame:** $S_0 = OXYZ = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, with $\mathbf{j}$ upwards.
- **Moving frame** (requested in (b)): $S_1 = OX_1Y_1Z_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the wire, with $OX_1$ in the plane of the wire (the $x$ axis of the figure), $OY_1 \equiv OY$ and $OZ_1$ normal to the plane of the wire. Taking $S_1$ coincident with $S_0$ at $t = 0$, it is a simple rotation of angle $\Omega t$ about $\mathbf{j}$ (theory note, Sec. 4.1):

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i} - \sin\Omega t\,\mathbf{k}, \qquad \mathbf{j}_1 = \mathbf{j}, \qquad \mathbf{k}_1 = \sin\Omega t\,\mathbf{i} + \cos\Omega t\,\mathbf{k}$$

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & 0 & \sin\Omega t \\ 0 & 1 & 0 \\ -\sin\Omega t & 0 & \cos\Omega t \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\Omega t + \sin^2\Omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

(the columns are unit vectors and $\mathbf{i}_1 \cdot \mathbf{k}_1 = \cos\Omega t \sin\Omega t - \sin\Omega t \cos\Omega t = 0$; also $\mathbf{i}_1 \times \mathbf{j}_1 = \mathbf{k}_1$).

- **Motion of $S_1$: $\mathbf{r}_0^O = \mathbf{0}$, $\mathbf{v}_0^O = \mathbf{0}$, $\mathbf{a}_0^O = \mathbf{0}$** (fixed origin); $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_1$, $\boldsymbol{\alpha}_{10} = \dot{\Omega}\,\mathbf{j}_1 = \mathbf{0}$. Check: $\left.\frac{d\mathbf{i}_1}{dt}\right|_0 = -\Omega\sin\Omega t\,\mathbf{i} - \Omega\cos\Omega t\,\mathbf{k} = -\Omega\,\mathbf{k}_1$ (chain rule), and $\Omega\,\mathbf{j} \times \mathbf{i}_1 = -\Omega\,\mathbf{k}_1$.
- **Position of $P$ in $S_1$:** with $\theta$ the angle of $CP$ measured from the downward vertical, positive towards $+X_1$, and $\theta \in [-\pi/2, \pi/2]$ on the wire:

$$\mathbf{r}_1^P = a\sin\theta\,\mathbf{i}_1 + a\,(1 - \cos\theta)\,\mathbf{j}_1$$

- **Wire basis (intrinsic frame of the circle):** tangent $\mathbf{e}_t = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1$, principal normal towards $C$, $\mathbf{e}_n = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1$, binormal $\mathbf{e}_t \times \mathbf{e}_n = (\cos^2\theta + \sin^2\theta)\,\mathbf{k}_1 = \mathbf{k}_1$. The matrix with columns $\mathbf{e}_t, \mathbf{e}_n, \mathbf{k}_1$ in $\mathcal{B}_1$ is a rotation by $\theta$ about $\mathbf{k}_1$, with determinant $\cos^2\theta + \sin^2\theta = 1$.

PHASE 3

### Step-by-Step Mathematical Deduction

**(a) Degrees of freedom**

**Pedagogical Justification:** CDOF = (coordinates of the free particle) − (independent holonomic constraints).

$P$ has 3 coordinates. In $S_1$ the wire imposes two fixed equations, $z_1 = 0$ (plane of the wire) and $x_1^2 + (y_1 - a)^2 = a^2$ (circle). In $S_0$ the same constraints are rheonomic (they turn with $\Omega t$ ), but the count is the same: CDOF $= 3 - 2 = 1$, with generalized coordinate $\theta$. The rotation $\Omega t$ is prescribed, so it is not a degree of freedom.

**(b) Inertia force in $OX_1Y_1Z_1$**

**Pedagogical Justification:** Notes Eq. 4.62 (Slide 28) gives $\mathbf{F}_i = -m\mathbf{a}_0^O - m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P - m\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) - 2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$. The first two terms vanish ($\mathbf{a}_0^O = \mathbf{0}, \boldsymbol{\alpha}_{10} = \mathbf{0}$).

**Relative velocity.** In $S_1$ only the components change; chain rule $\frac{d}{dt}\sin\theta = \cos\theta\,\dot\theta$, $\frac{d}{dt}(1 - \cos\theta) = \sin\theta\,\dot\theta$:

$$\mathbf{v}_1^P = a\dot\theta\cos\theta\,\mathbf{i}_1 + a\dot\theta\sin\theta\,\mathbf{j}_1 = a\dot\theta\,\mathbf{e}_t$$

**Centrifugal force.** With $\mathbf{j}_1 \times \mathbf{i}_1 = -\mathbf{k}_1$ and $\mathbf{j}_1 \times \mathbf{j}_1 = \mathbf{0}$, $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \Omega\,\mathbf{j}_1 \times a\sin\theta\,\mathbf{i}_1 = -\Omega a\sin\theta\,\mathbf{k}_1$; then, with $\mathbf{j}_1 \times \mathbf{k}_1 = \mathbf{i}_1$, $\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) = -\Omega^2 a\sin\theta\,\mathbf{i}_1$:

$$\mathbf{F}_C = m\Omega^2 a\sin\theta\,\mathbf{i}_1$$

It points horizontally away from the axis and equals $m\Omega^2\mathbf{r}_\perp$ (Slide 30), since $\mathbf{r}_\perp = a\sin\theta\,\mathbf{i}_1$ is the part of $\mathbf{r}_1^P$ perpendicular to $\mathbf{j}_1$.

**Coriolis force.** $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = \Omega\,\mathbf{j}_1 \times \left(a\dot\theta\cos\theta\,\mathbf{i}_1 + a\dot\theta\sin\theta\,\mathbf{j}_1\right) = -\Omega a\dot\theta\cos\theta\,\mathbf{k}_1$, so

$$\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = 2m\Omega a\dot{\theta}\cos\theta\,\mathbf{k}_1$$

**Total inertia force:**

$$\mathbf{F}_i = m\Omega^2 a\sin\theta\,\mathbf{i}_1 + 2m\Omega a\dot{\theta}\cos\theta\,\mathbf{k}_1$$

**(c) Weight in $OX_1Y_1Z_1$**

The axis of rotation is the vertical, so $\mathbf{j}_1 = \mathbf{j}$ at all times and

$$\mathbf{W} = -mg\,\mathbf{j}_1 \qquad (W_{X_1} = 0,\quad W_{Y_1} = -mg,\quad W_{Z_1} = 0)$$

In the wire basis: $\mathbf{W}\cdot\mathbf{e}_t = -mg\sin\theta$, $\mathbf{W}\cdot\mathbf{e}_n = -mg\cos\theta$, $\mathbf{W}\cdot\mathbf{k}_1 = 0$.

**(d) Equilibrium points**

> **Pedagogical Justification:** Newton's law in $S_1$, $m\mathbf{a}_1^P = \mathbf{W} + \mathbf{N} + \mathbf{F}_i$ (Notes Eq. 4.63), projected on the tangent $\mathbf{e}_t$, eliminates the smooth reaction ($\mathbf{N}\cdot\mathbf{e}_t = 0$) and the Coriolis force ($\mathbf{k}_1\cdot\mathbf{e}_t = 0$).

**Relative acceleration.** Product rule on $a\dot{\theta}\,\mathbf{e}_t$ in $S_1$, with $\left.\frac{d\mathbf{e}_t}{dt}\right|_1 = \dot{\theta}\,(-\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1) = \dot{\theta}\,\mathbf{e}_n$:

$$\mathbf{a}_1^P = a\ddot{\theta}\,\mathbf{e}_t + a\dot{\theta}^2\,\mathbf{e}_n$$

**Tangential equation.** $\mathbf{F}_C\cdot\mathbf{e}_t = m\Omega^2 a\sin\theta\cos\theta$, so

$$ma\ddot{\theta} = -mg\sin\theta + m\Omega^2 a\sin\theta\cos\theta \;\Rightarrow\; \ddot{\theta} = \sin\theta\left(\Omega^2\cos\theta - \frac{g}{a}\right)$$

**Equilibria** (relative rest, $\ddot{\theta} = \dot{\theta} = 0$):

1. $\sin\theta = 0$: $\theta = 0$, the lowest point $O$. It exists for every $\Omega$.

2. $\cos\theta_* = \dfrac{g}{a\Omega^2}$: two symmetric points $\theta = \pm\theta_*$, which exist only if $\Omega^2 > g/a$ (then $0 < \theta_* < \pi/2$, inside the wire).

**Stability** with the potential of (f), $V_1(\theta) = mga(1-\cos\theta) - \frac{1}{2}m\Omega^2 a^2 \sin^2\theta$ (the tangential equation is $ma^2\ddot\theta = -dV_1/d\theta$). Its derivatives are $V_1' = ma\sin\theta\left(g - \Omega^2 a\cos\theta\right)$ and $V_1'' = mga\cos\theta - m\Omega^2 a^2\cos 2\theta$:

- $V_1''(0) = ma\left(g - \Omega^2 a\right)$: $\theta = 0$ is stable if $\Omega^2 < g/a$ and unstable if $\Omega^2 > g/a$.
- $V_1''(\pm\theta_*) = mga\cos\theta_* - m\Omega^2 a^2\left(2\cos^2\theta_* - 1\right) = m\Omega^2 a^2\left(1 - \cos^2\theta_*\right) = m\Omega^2 a^2\sin^2\theta_* > 0$ (using $ga = \Omega^2 a^2\cos\theta_*$): the off-axis equilibria are stable whenever they exist.

**(e) Quadratures and reaction force**

> **Pedagogical Justification:** multiplying the tangential equation by $\dot\theta$ makes both sides exact derivatives; the result is the energy integral of (f). The remaining two projections of Newton's law (on $\mathbf{e}_n$ and $\mathbf{k}_1$) give the reaction.

**First integral.** Multiply $ma^2\ddot\theta = -mga\sin\theta + m\Omega^2 a^2\sin\theta\cos\theta$ by $\dot\theta$. By the chain rule, $\frac{d}{dt}\left(\frac{1}{2}\dot\theta^2\right) = \dot\theta\ddot\theta$, $\frac{d}{dt}\cos\theta = -\sin\theta\,\dot\theta$ and $\frac{d}{dt}\left(\frac{1}{2}\sin^2\theta\right) = \sin\theta\cos\theta\,\dot\theta$, so

$$\frac{d}{dt}\left[\tfrac{1}{2}ma^2\dot\theta^2 - mga\cos\theta - \tfrac{1}{2}m\Omega^2 a^2\sin^2\theta\right] = 0$$

Integrate from $0$ to $t$ and apply Barrow's rule, with $\theta(0) = 0$ and $a\dot\theta(0) = v_0$ (at $O$, $\mathbf{r}_1^P = \mathbf{0}$, so $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \mathbf{0}$ and the given absolute velocity equals the relative one, $v_0\,\mathbf{i}_1 = a\dot\theta(0)\,\mathbf{e}_t(0)$):

$$\left[\tfrac{1}{2}ma^2\dot\theta^2 - mga\cos\theta - \tfrac{1}{2}m\Omega^2 a^2\sin^2\theta\right]_0^t = 0 \;\Rightarrow\; \dot\theta^2 = \frac{v_0^2}{a^2} - \frac{2g}{a}\left(1-\cos\theta\right) + \Omega^2\sin^2\theta \equiv \Phi(\theta)$$

**Quadrature.** Separate variables ($d\theta = \sqrt{\Phi}\,dt$, positive root since $\dot\theta(0) = v_0/a > 0$, valid until the first turning point $\Phi = 0$) and integrate from $0$ to $t$:

$$t = \int_0^{\theta(t)} \frac{d\theta'}{\sqrt{\dfrac{v_0^2}{a^2} - \dfrac{2g}{a}\left(1-\cos\theta'\right) + \Omega^2\sin^2\theta'}}$$

**Reaction force.** $\mathbf{N} = N_n\,\mathbf{e}_n + N_z\,\mathbf{k}_1$ (no tangential component). Projection on $\mathbf{e}_n$, with $\mathbf{F}_C \cdot \mathbf{e}_n = -m\Omega^2 a \sin^2\theta$:

$$ma\dot\theta^2 = N_n - mg\cos\theta - m\Omega^2 a\sin^2\theta \;\Rightarrow\; N_n = ma\dot\theta^2 + mg\cos\theta + m\Omega^2 a\sin^2\theta$$

Substituting $\dot\theta^2 = \Phi(\theta)$:

$$N_n(\theta) = \frac{mv_0^2}{a} + mg\,(3\cos\theta - 2) + 2m\Omega^2 a\sin^2\theta$$

Projection on $\mathbf{k}_1$ ($\mathbf{a}_1^P \cdot \mathbf{k}_1 = 0$, $\mathbf{W}\cdot\mathbf{k}_1 = 0$):

$$0 = N_z + 2m\Omega a\dot\theta\cos\theta \;\Rightarrow\; N_z = -2m\Omega a\dot\theta\cos\theta = \mp 2m\Omega a\cos\theta\sqrt{\Phi(\theta)}$$

(upper sign while $\dot\theta > 0$). The total reaction is $\mathbf{N} = N_n\,(-\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1) + N_z\,\mathbf{k}_1$.

**Staying on the wire.** The particle reaches the end $\theta = \pi/2$ only if $\Phi(\pi/2) = v_0^2/a^2 - 2g/a + \Omega^2 \geq 0$; otherwise it oscillates between the two roots of $\Phi(\theta) = 0$.

**(f) Energy equation**

> **Pedagogical Justification:** $E_1^P$ is conserved when every force that does work in $S_1$ derives from a time-independent potential (Slide 29; theory note, Sec. 12.6).

**Constant $\Omega$: yes.** In $S_1$: the wire is fixed and smooth, so $\mathbf{N}\cdot\mathbf{v}_1^P = 0$; the Coriolis force never does work, $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ (Slide 31); the weight has the potential $mgy_1 = mga(1-\cos\theta)$; the centrifugal force has the potential $-\tfrac12 m\Omega^2 r_\perp^2 = -\tfrac12 m\Omega^2 a^2\sin^2\theta$ (Slide 30), time-independent because $\Omega$ is constant. Hence

$$E_1^P = \tfrac12 ma^2\dot\theta^2 + mga\,(1-\cos\theta) - \tfrac12 m\Omega^2 a^2\sin^2\theta = \tfrac12 mv_0^2$$

which is the first integral used in (e). (In $S_0$ the mechanical energy is not conserved: the reaction $N_z$ is perpendicular to the moving wire but not to the absolute velocity, so the motor that keeps $\Omega$ constant does work.)

**Variable $\Omega(t)$: no energy integral.** Two things change. The Euler force appears, $\mathbf{F}_E = -m\dot{\Omega}\,\mathbf{j}_1 \times a\sin\theta\,\mathbf{i}_1 = m\dot{\Omega}a\sin\theta\,\mathbf{k}_1$; it is normal to the plane of the wire, so $\mathbf{F}_E \cdot \mathbf{v}_1^P = 0$ and it does no work in $S_1$ (it is balanced by $N_z$). But the centrifugal potential now depends explicitly on time, $V_C = -\frac{1}{2}m\Omega(t)^2 a^2 \sin^2\theta$, and differentiating $E_1^P$ along the motion (tangential equation with $\Omega(t)$) gives

$$\frac{dE_1^P}{dt} = \frac{\partial V_C}{\partial t} = -m\Omega\dot{\Omega}a^2\sin^2\theta \neq 0$$

so $E_1^P$ is not a constant of motion and the problem cannot be reduced to a quadrature with the energy equation. The work-energy relation $dE_1^P/dt = \partial V_C/\partial t$ is still valid, but it is not a first integral.

> **Correction to the Topic 3 solution**
> The Topic 3 version of this problem placed the origin at the centre of the circle (the statement puts it at the lowest point $O$), gave only one off-axis equilibrium (there are two, $\pm\theta_*$), gave only the out-of-plane reaction (the in-plane component $N_n$ is also part of the answer) and said that for variable $\Omega$ "Euler forces do work". The Euler force $m\dot{\Omega}a\sin\theta\,\mathbf{k}_1$ is perpendicular to $\mathbf{v}_1^P$ and does no work; the energy integral is lost because the centrifugal potential depends explicitly on time. The quadrature $\dot{\theta}^2 = \Phi(\theta)$, the equilibrium $\cos\theta_* = g/(a\Omega^2)$ and $N_z = -2m\Omega a\dot{\theta}\cos\theta$ of the Topic 3 version agree with this solution.

PHASE 4

### Physical Interpretation & Dimensional Verification

- **Bifurcation.** For $\Omega^2 < g/a$ gravity dominates and the bottom is the only (stable) equilibrium. At $\Omega_c = \sqrt{g/a}$ the bottom becomes unstable and two stable equilibria $\pm\theta_*$ appear, where the tangential components of weight and centrifugal force balance: $mg\sin\theta_* = m\Omega^2 a\sin\theta_*\cos\theta_*$.
- **Limits.** $\Omega \to 0$: $\Phi = v_0^2/a^2 - 2g(1-\cos\theta)/a$ and $N_n = mv_0^2/a + mg(3\cos\theta - 2)$, the simple-pendulum results; $N_z \to 0$. $\Omega \to \infty$: $\cos\theta_* \to 0$, the equilibria move to the ends of the wire.

- **Coriolis and the reaction.** $N_z$ is exactly the force the wire must exert to cancel the Coriolis force; it changes sign with $\dot\theta$, pushing the particle forward or backward with the rotation.
- **Dimensions:** $[\Omega^2] = [g/a] = \mathrm{s}^{-2}$, so every term of $\Phi$ is in $\mathrm{s}^{-2}$ and $[t] = \mathrm{rad}/\mathrm{s}^{-1} = \mathrm{s}$; $[mv_0^2/a] = [mg] = [m\Omega^2 a] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$; $[m\Omega a\dot\theta] = \mathrm{N}$; $[E_1^P] = [ma^2\dot\theta^2] = \mathrm{J}$.
- **Symbolic check (sympy, `verify_P26.py`):** $[{}_0R_1]$ and $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}$; Notes Eq. 3.27 against the double derivative of $\mathbf{r}_1^P$ in $\mathcal{B}_0$; $\mathbf{F}_C, \mathbf{F}_{COR}$; the equation of motion and both reaction components from $m\mathbf{a}_0^P = \mathbf{W} + \mathbf{N}$ in $S_0$; $dE_1^P/dt = 0$; $\Phi(\theta)$ and $N_n(\theta)$; the stability derivatives; and, for $\Omega(t)$, the Euler force, its zero power and $dE_1^P/dt = -m\Omega\dot\Omega a^2\sin^2\theta$. A numerical integration ($g = 9.81$, $a = 1$, $\Omega = 4$, $v_0 = 2$, SI) keeps $\dot\theta^2 - \Phi(\theta)$ below $3\times 10^{-10}$.

---

*Source: Problems.pdf (PDF page 14)*